



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
(An Institution of National Importance under the Ministry of Education, Govt. of India)
MID SEMESTER EXAMINATIONS-II
M&C IV Semester

MAPC-206: Theory of Computation

Time: 04:00 PM to 04:50 PM

Max. Marks: 15

Date: 07-04-2026

N.B.: Answer ALL questions. Each question carries equal marks.
Answers to all parts of each question should be in one place.

Q1	Design a Mealy Machine accepting the language consisting of strings from Σ^* , where $\Sigma = \{a, b\}$ and the strings should end with either aa or bb .
Q2	Define the Pumping Lemma and write its application by showing one example of your choice.
Q3	Construct a PDA that accepts even palindromes including epsilon over $\Sigma = \{a, b\}$. Explain the stack analysis.